Income Distribution and Macroeconomics

Oded Galor and Joseph Zeira

Review of Economic Studies, 1993

Fundamental Questions

- What role does heterogeneity play in macroeconomics?
- Does income distribution affect aggregate economic activity?
- Does inequality foster or hamper economic growth?

The Traditional Perspective

- Inequality promotes economic growth

  Keynes (1919), Kaldor (1957)

  - The marginal propensity to save increases with income
  - Inequality channels resources towards individuals whose marginal propensity to save is higher
    - Inequality $\Rightarrow$ Capital accumulation $\Rightarrow$ Economic growth

The Representative-Agent Paradigm

- Income distribution has no effect on macroeconomic activity and economic growth
  - The representative-agent approach rests on the premise that heterogeneity and income distribution do not affect aggregate economic outcomes
  - Aggregate economic activity is determined independently of the distribution of income and wealth
    - Growth Process $\Rightarrow$ Factor Incomes & income distribution
    - Income Distribution $\not\Rightarrow$ Growth Process

The Modern Perspective: Origins

- A Heterogeneous-Agent Model

  Galor and Zeira (1988, 1993)

  - Unlike the Representative-Agent Paradigm
    - Income distribution affects aggregate economic activity and growth
  - Unlike the Traditional Perspective
    - Inequality can hamper human capital formation and economic growth

The Galor-Zeira Model

- An overlapping-generations economy
- Discrete time: $t = 0, 1, 2, \ldots$
- One homogeneous good
- Three factors of production
  - K: Physical capital
  - L^s: Skilled labor
  - L^u: Unskilled labor

Production

- Output produced in period t

$$Y_t = Y_t^s + Y_t^u$$

  - Production in the skilled-intensive sector

$$Y_t^s = F(K_t, L_t^s) = L_t^s f(k_t), \qquad k_t \equiv K_t / L_t^s$$

    - CRS, $f'(k) > 0$ and $f''(k) < 0$

  - Production in the unskilled-intensive sector

$$Y_t^u = aL_t^u, \qquad a > 0$$

Inverse Demand for Factors

- Capital:

$$r_t = f'(k_t) \equiv r(k_t)$$

- Skilled labor:

$$w_t^s = f(k_t) - f'(k_t)k_t \equiv w^s(k_t)$$

- Unskilled labor:

$$w_t^u = a \equiv w^u$$

Factor Prices

- A small open economy with access to international capital markets
- Firms can borrow at the constant world interest rate $r > 0$
  - Factor prices

$$r_t = r$$

$$k_t = (f')^{-1}(r) \equiv k$$

$$w_t^s = w^s(k) \equiv w^s$$

  - $\Rightarrow$ Factor prices are constant over time

$$\boxed{(r_t, w_t^s, w_t^u) = (r, w^s, w^u) \qquad \forall t}$$

Individuals

- A continuum of measure one in each generation
- Each individual has one parent and one child
  - Individuals are Identical in:
    - Preferences
    - Innate abilities
  - Individuals differ in
    - Inherited wealth $\Rightarrow$ Investment in human capital

Member of Generation t

- Lives for two periods: $(t, t+1)$
  - Period t (First period of life):
    - Invests in human capital or works as an unskilled worker
  - Period $t+1$ (Second period of life):
    - Works as a skilled worker if educated in period t
    - Works as an unskilled worker otherwise

Member of Generation t: Endowments and Preferences

- Endowments:
  - 1 unit of time in each period
  - Inherited wealth b_t received in the first period
- Preferences: "Joy of Giving"

$$u^t = \alpha \ln c_{t+1} + (1-\alpha) \ln b_{t+1}, \qquad \alpha \in (0,1)$$

  - c_{t+1}: Household's consumption in the second period (including child's)
  - b_{t+1}: Bequest to offspring

Member of Generation t: Budget Constraint

- Second-period budget constraint

$$c_{t+1} + b_{t+1} \leq \omega_{t+1}$$

  - c_{t+1}: Consumption in the second period
  - b_{t+1}: Bequest to offspring
  - ω_{t+1}: Wealth in the second period

Member of Generation t: Optimization Problem

- Choice of consumption and bequest

$$(c_{t+1}, b_{t+1}) = \arg\max \{\alpha \ln c_{t+1} + (1-\alpha) \ln b_{t+1}\}$$

$$\text{subject to} \qquad c_{t+1} + b_{t+1} \leq \omega_{t+1}$$

$$c_{t+1} > 0, \qquad b_{t+1} > 0$$

Member of Generation t: Optimization

- Consumption and bequest:

$$c_{t+1} = \alpha\omega_{t+1}$$

$$b_{t+1} = (1-\alpha)\omega_{t+1}$$

- Indirect Utility:

$$\begin{aligned} v^t &= \alpha \ln \alpha\omega_{t+1} + (1-\alpha)\ln[(1-\alpha)\omega_{t+1}] \\ &= [\alpha \ln \alpha + (1-\alpha)\ln(1-\alpha)] + \ln \omega_{t+1} \end{aligned}$$

$\Longrightarrow$ v^t is strictly increasing in 2nd period wealth, ω_{t+1}

$\Longrightarrow$ maximization of ω_{t+1} is the basis of occupational choices

Fundamental Departures from the Neoclassical Benchmark

- Imperfect capital markets

$$r < i \tag{A1}$$

  - $r \equiv$ Interest rate received by lenders
  - $i \equiv$ Interest rate paid by borrowers investing in human capital

- Non-convexities in the production of human capital
  - A fixed cost of education
  - Indivisibility of investment in human capital

$$h > 0 \tag{A2}$$

Wealth of an Individual Who Chooses an Unskilled Career

$$\begin{aligned} \omega_{t+1}^{u} &= (w^{u} + b_t)(1 + r) + w^{u} \\ &= w^{u}(2 + r) + (1 + r)b_t \end{aligned}$$

Wealth of an Individual Who Chooses a Skilled Career

$$\omega_{t+1}^{s} = \begin{cases} w^{s} - (h - b_t)(1+i) & \text{if} \quad b_t \leq h \\ \\ w^{s} + (b_t - h)(1+r) & \text{if} \quad b_t \geq h \end{cases}$$

$$\Longrightarrow$$

$$\omega_{t+1}^{s} = \begin{cases} w^{s} - (1+i)h + (1+i)b_t & \text{if} \quad b_t \leq h \\ \\ w^{s} - (1+r)h + (1+r)b_t & \text{if} \quad b_t \geq h \end{cases}$$

Modeling Assumptions: Co-Existence of Skilled and Unskilled Workers

- Modeling assumptions designed to ensure that occupational choices will result in some workers of each type
  - Investment in human capital is *not* beneficial for individuals who must finance the entire cost of education via borrowing

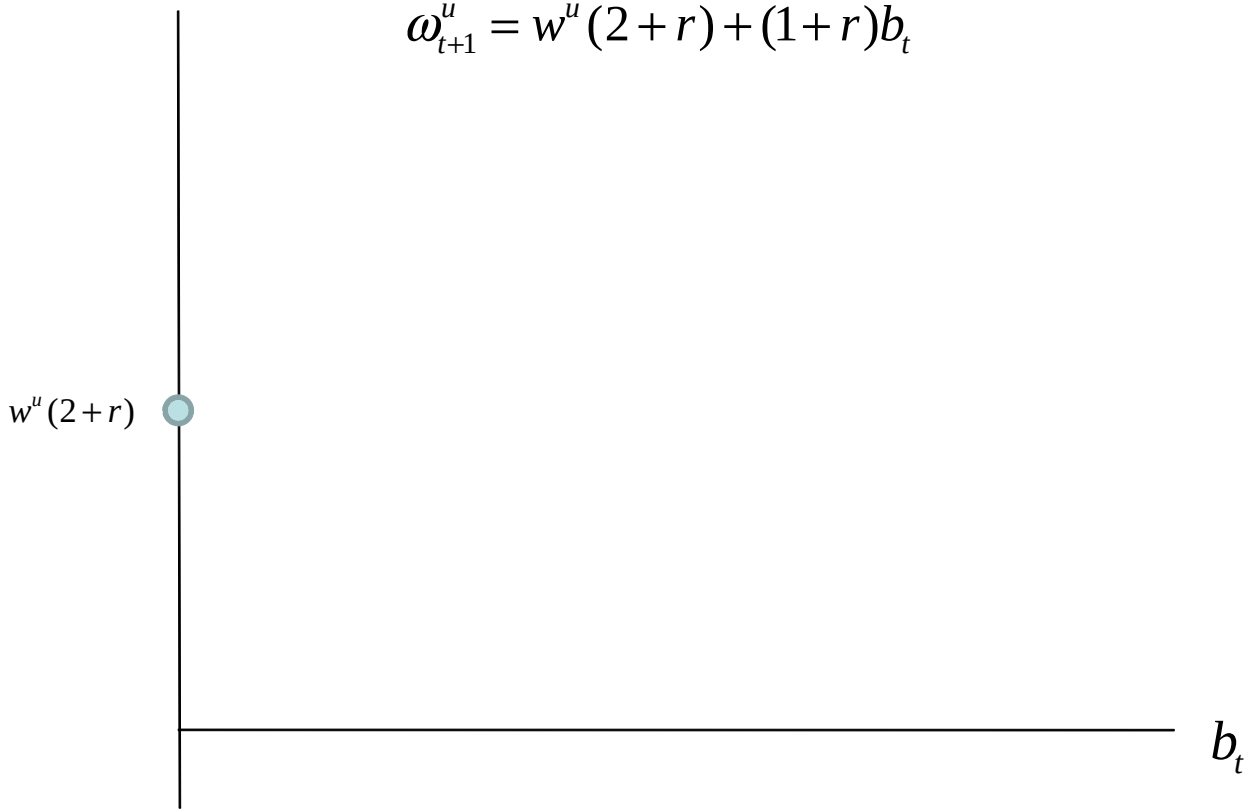
$$w^s - (1+i)h < 0 < w^u(2+r) \tag{A3}$$

  - Investment in human capital is beneficial for individuals who can finance the entire cost of education *without* borrowing

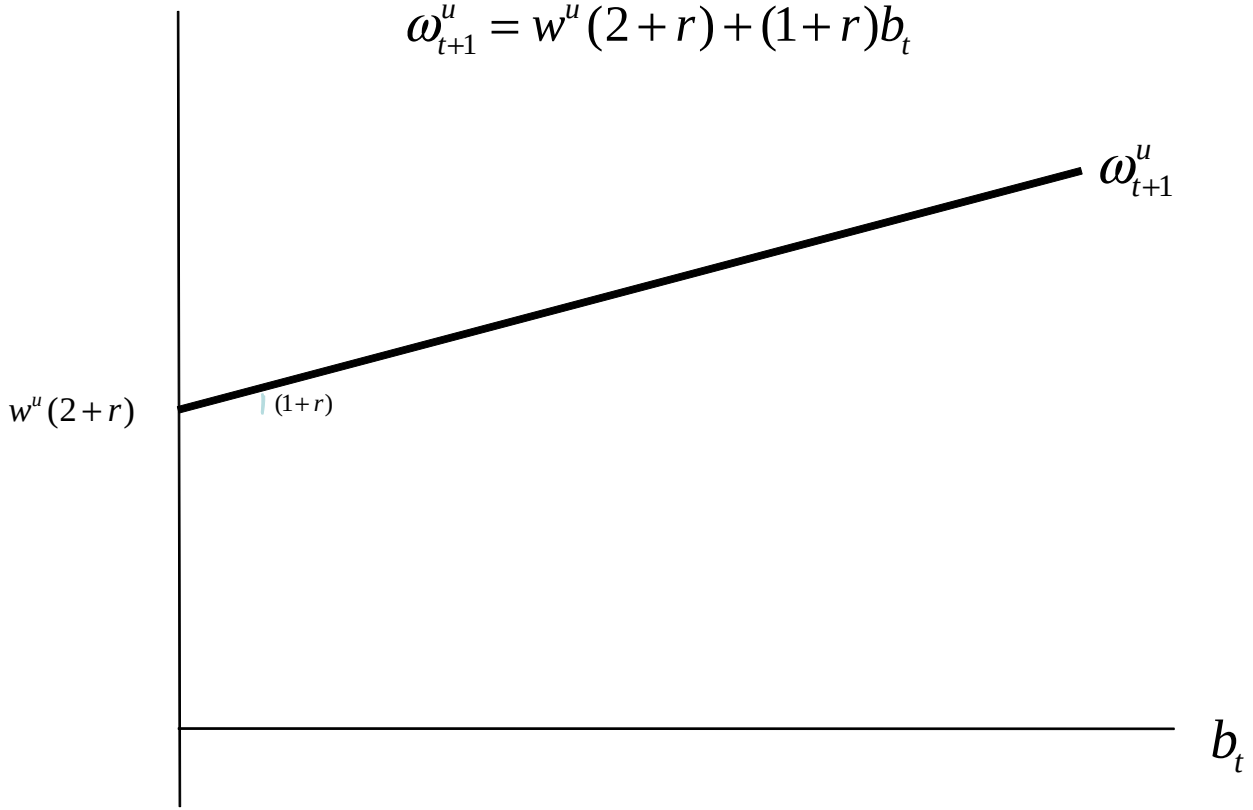
$$w^s - (1+r)h > w^u(2+r) \tag{A4}$$

Wealth from Being an Unskilled Worker

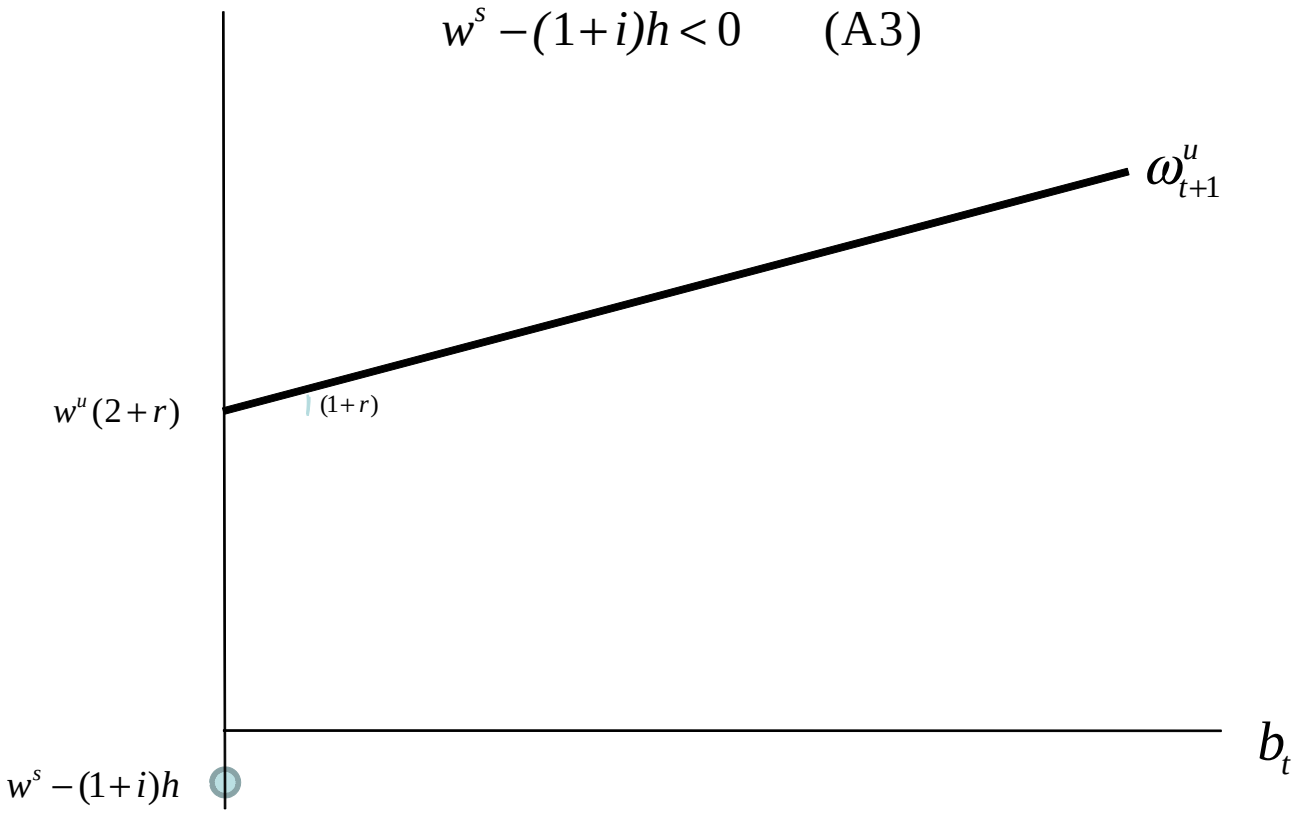
$$\omega^u_{t+1} = w^u(2+r)+(1+r)b_t$$



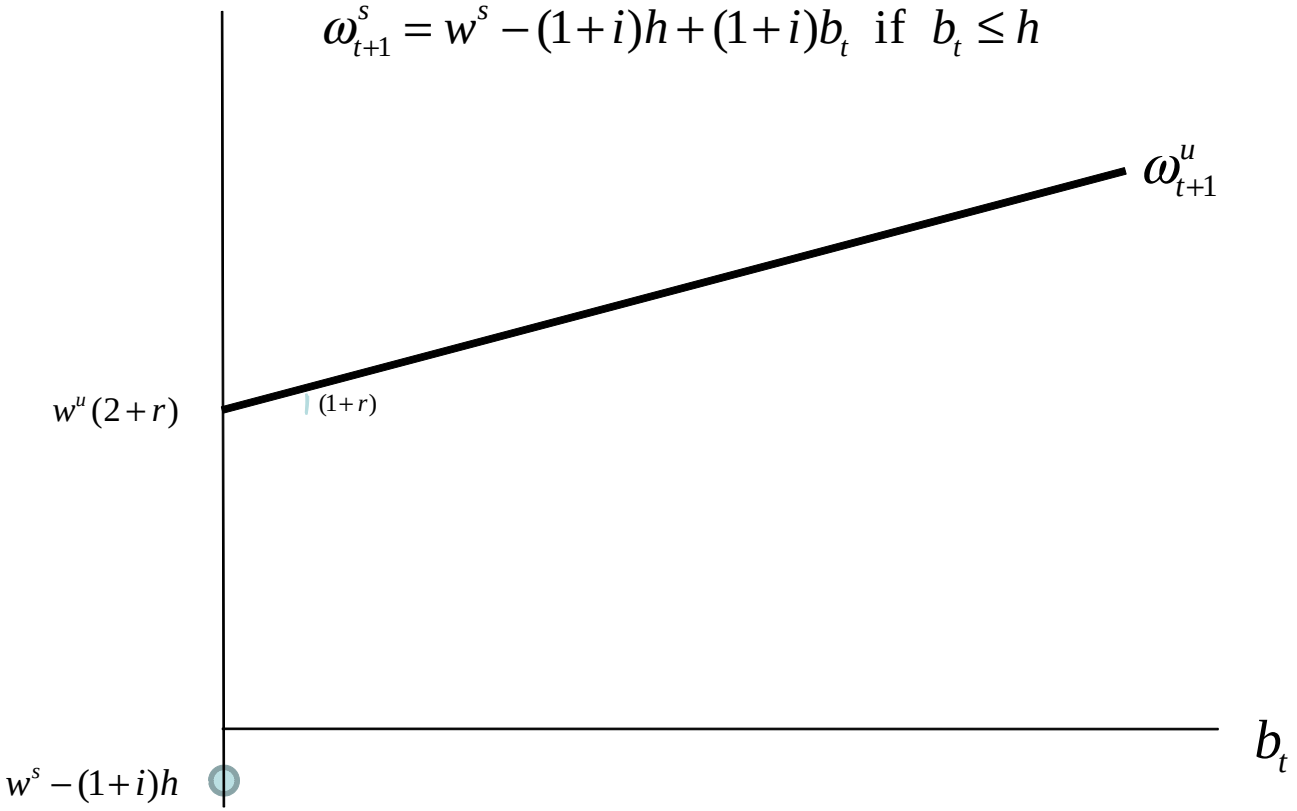
Wealth from Being an Unskilled Worker



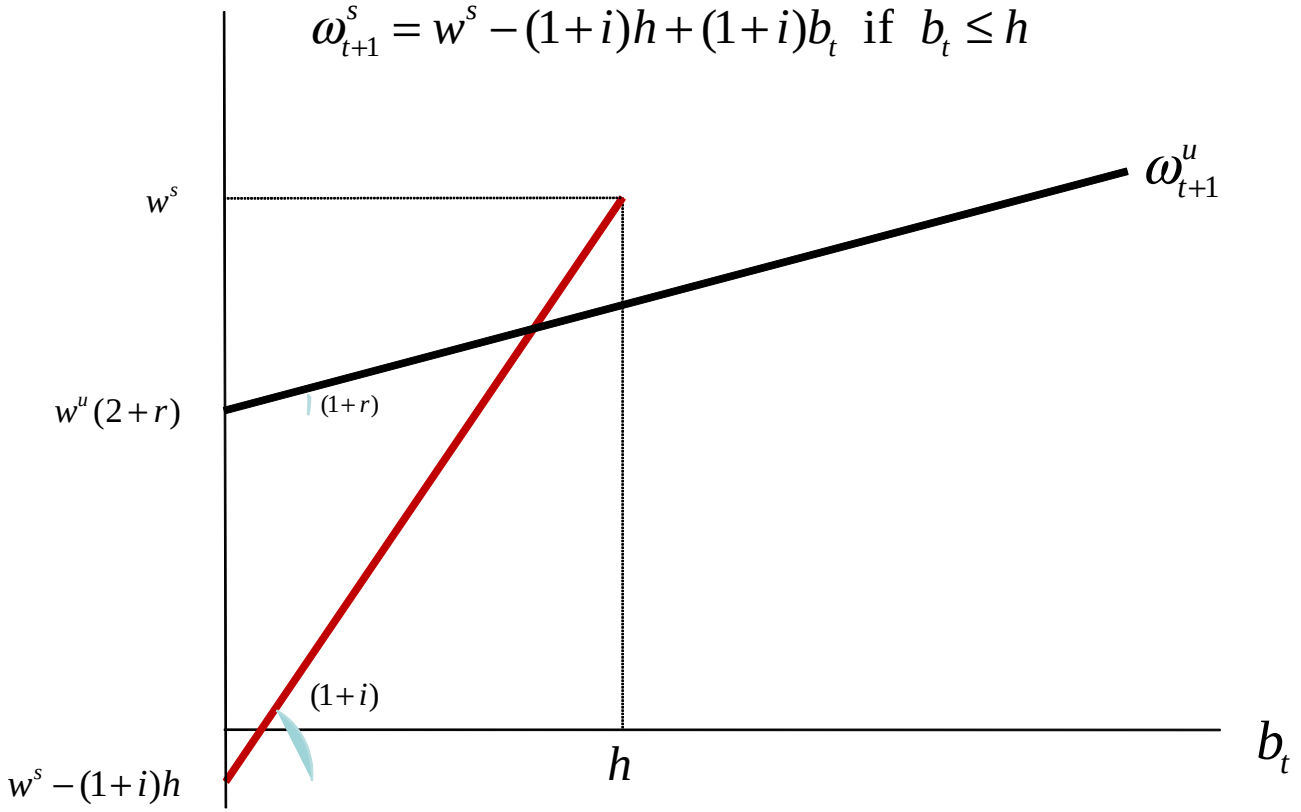
Wealth from Being a Skilled Worker: Borrowers



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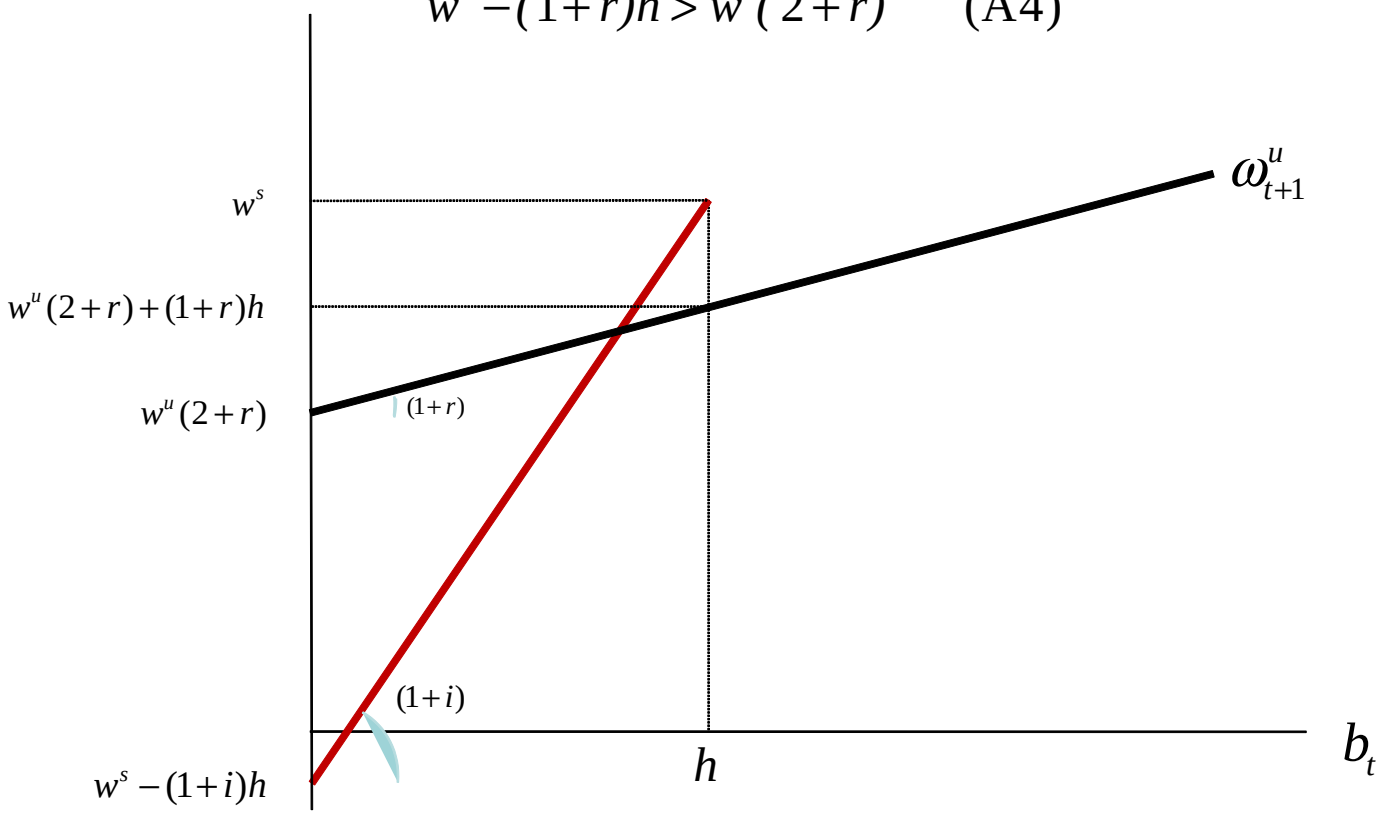


Wealth from Being a Skilled Worker: Borrowers

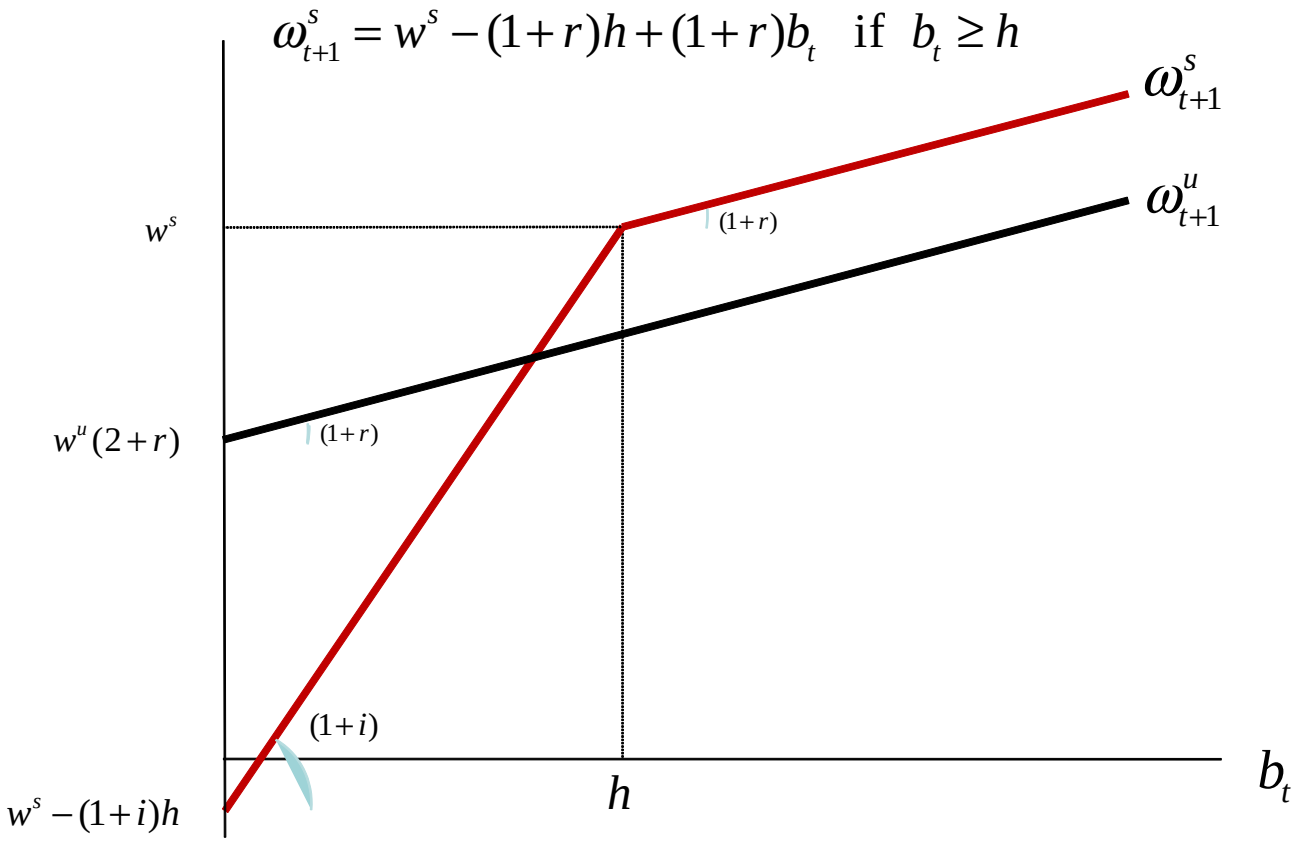


Wealth from Being a Skilled Worker: Borrowers

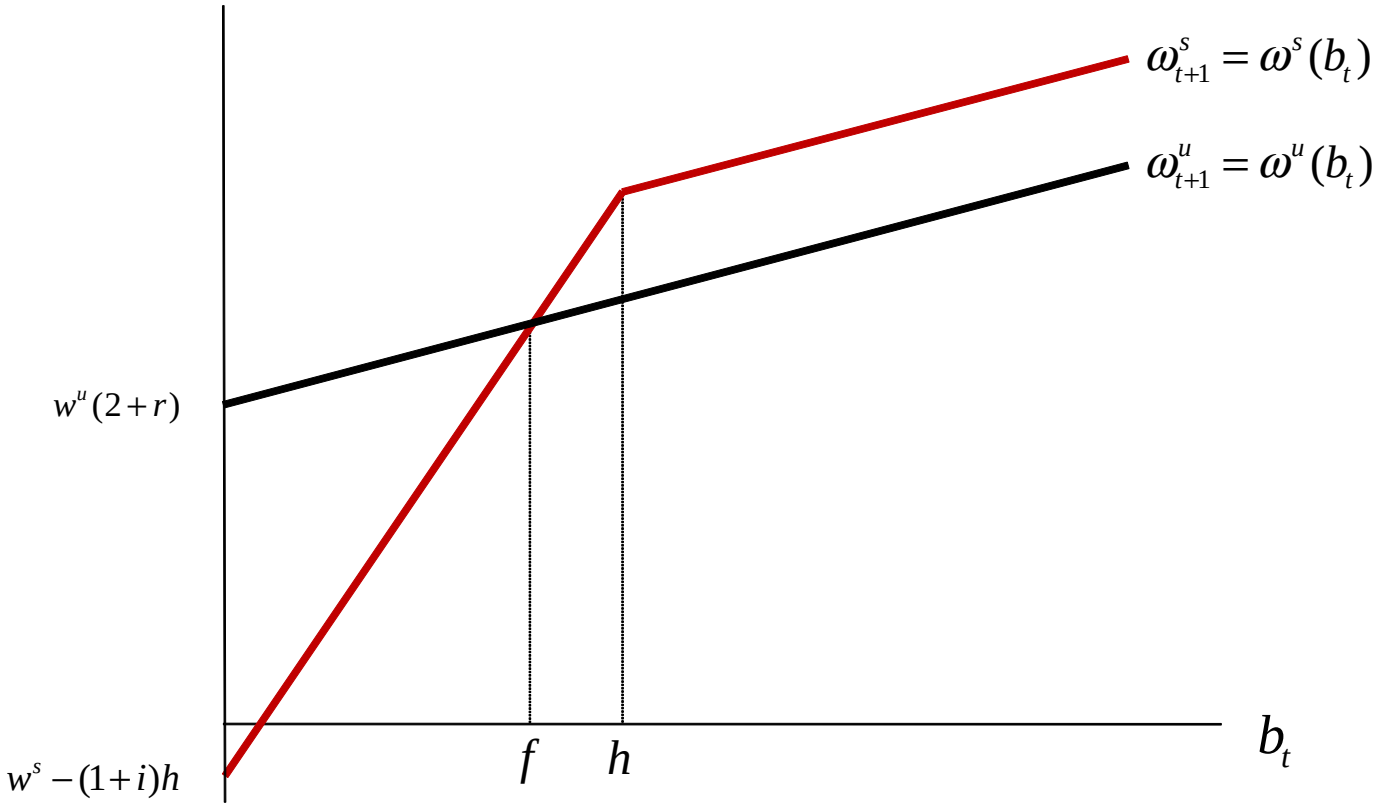
$$w^s - (1+r)h > w^u(2+r) \qquad \text{(A4)}$$



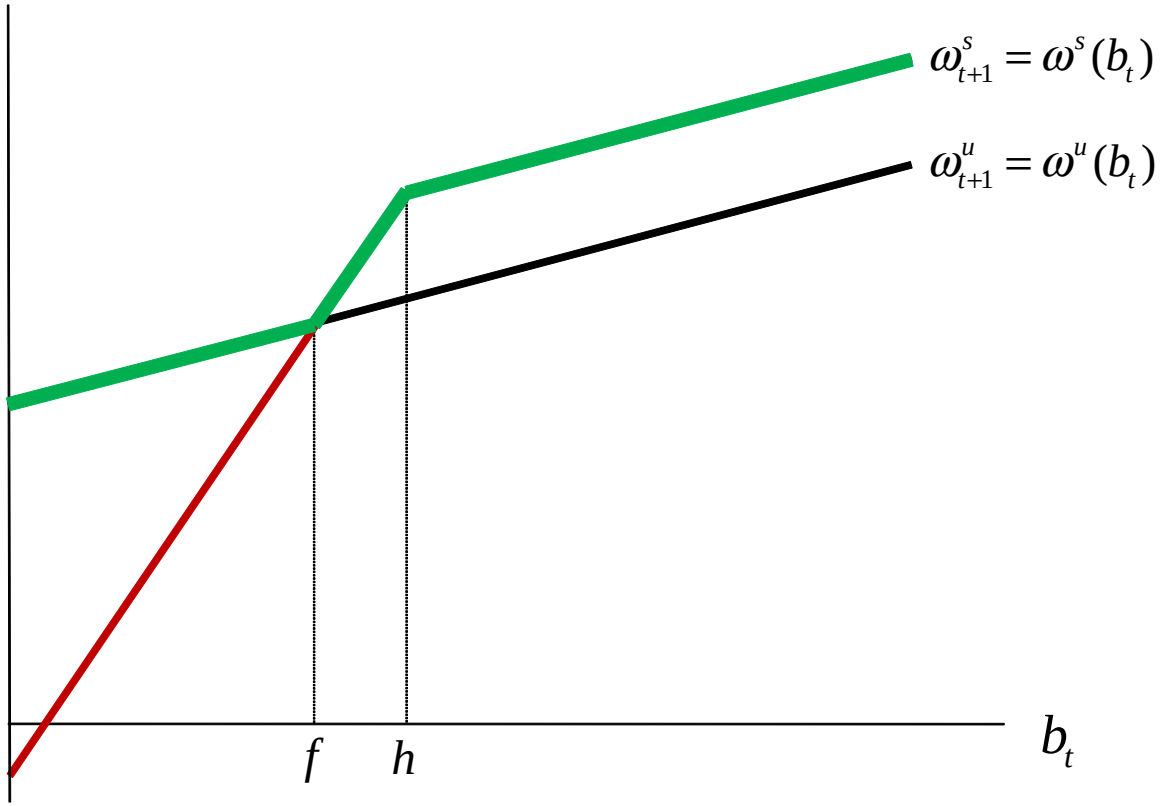
Wealth from Being a Skilled Worker: Lenders



Bequest and Occupational Choice



Bequest and Occupational Choice



Bequest and Occupational Choice

$$b_t \quad \begin{cases} < f \quad \rightarrow \omega^u_{t+1} > \omega^s_{t+1} \text{ (individual } t \text{ becomes unskilled)} \\ \\ > f \quad \rightarrow \omega^u_{t+1} < \omega^s_{t+1} \text{ (individual } t \text{ becomes skilled)} \end{cases}$$

where

$$f = \frac{w^u(2+r) - [w^s - (1+i)h]}{i - r} > 0$$

Bequest Dynamics

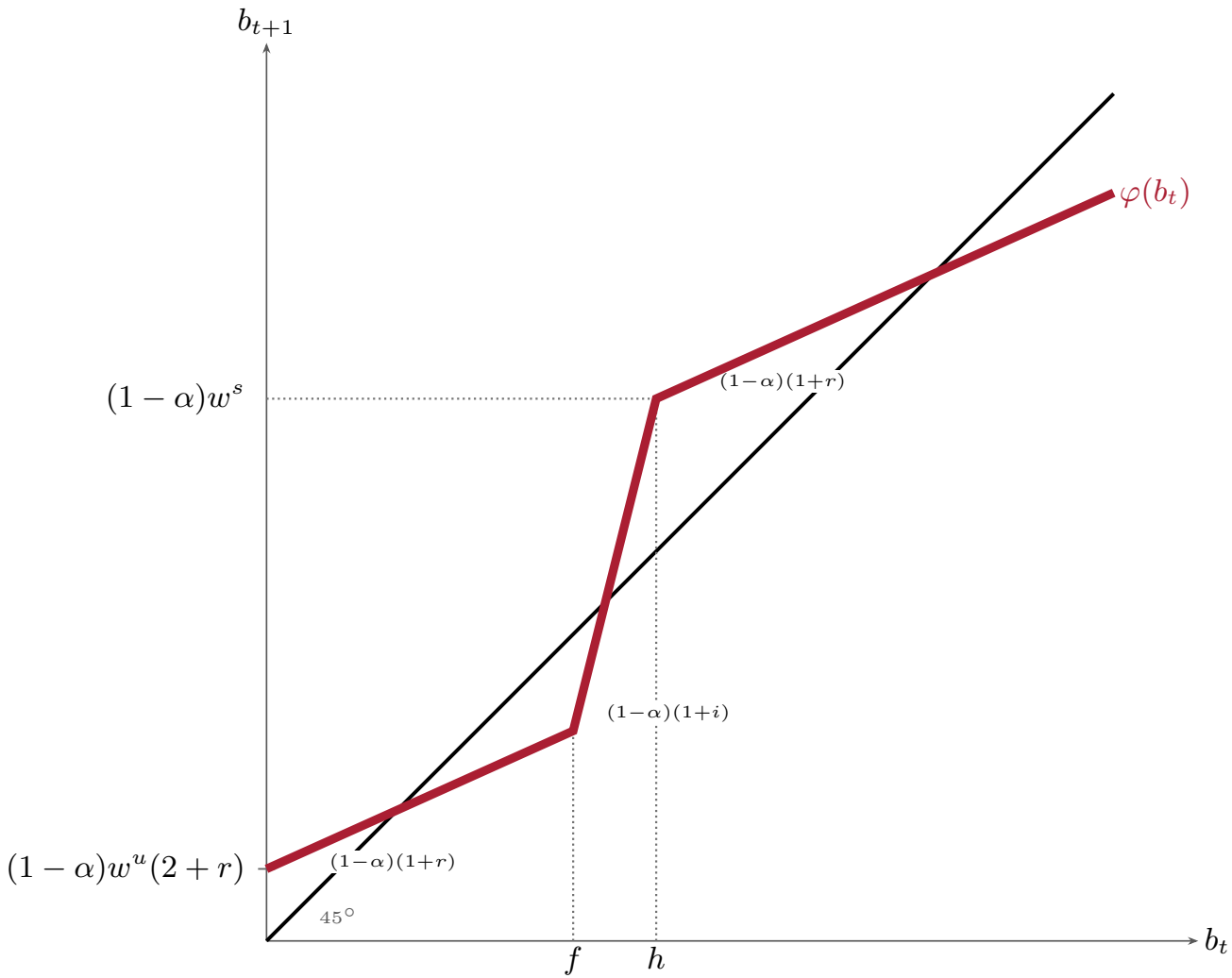
$$b_{t+1} = (1-\alpha)\omega_{t+1}$$

$$b_{t+1} = \begin{cases} (1-\alpha)\left[w^u(2+r) + (1+r)b_t\right] & b_t \in [0, f] \\ (1-\alpha)\left[w^s - (1+i)h + (1+i)b_t\right] & b_t \in [f, h] \\ (1-\alpha)\left[w^s - (1+r)h + (1+r)b_t\right] & b_t \in [h, \infty) \end{cases}$$

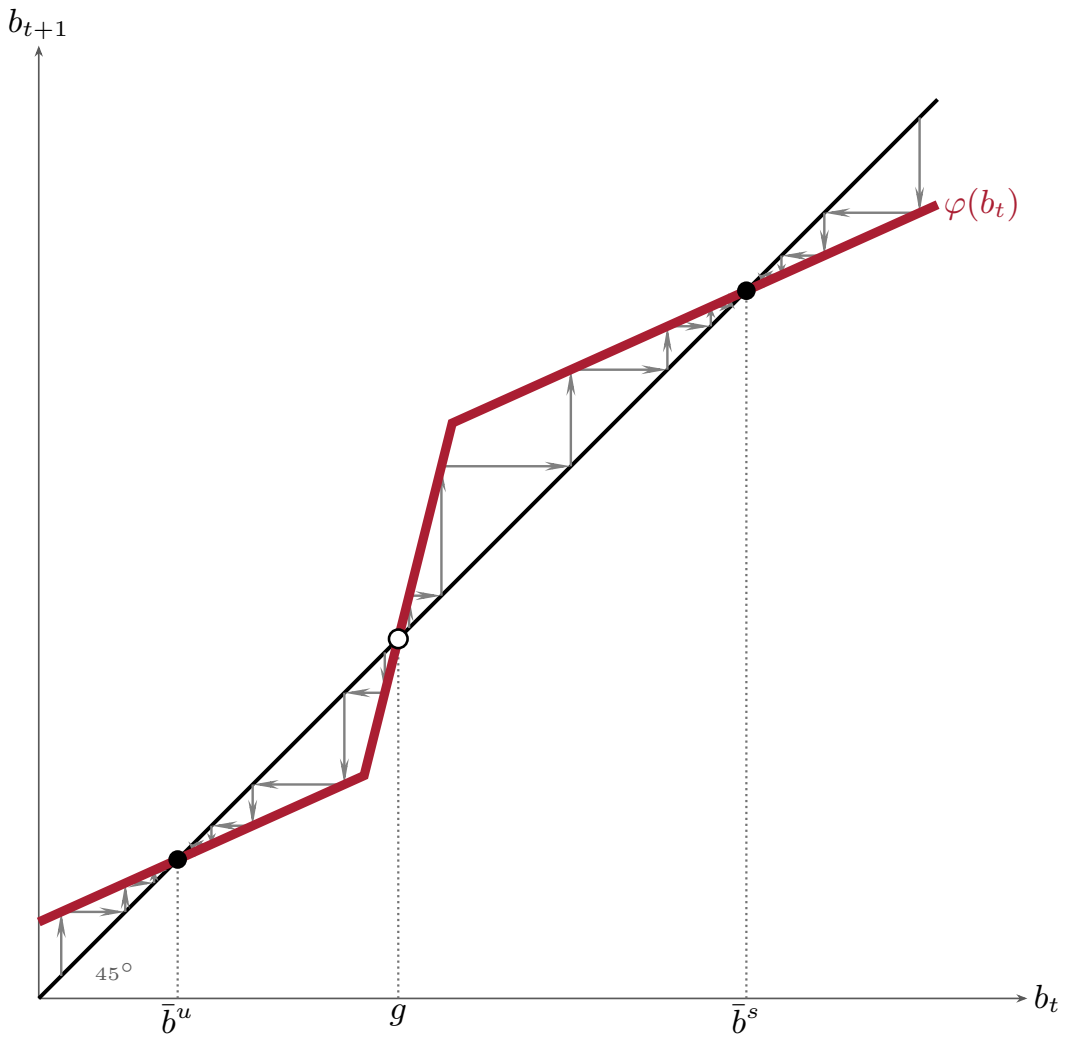
Path Dependence

- Sufficient Conditions for Multiple Steady States:
  - $(1-\alpha)(1+r)<1$
  - $(1-\alpha)(1+i)>1$
  - $\phi(f)=(1-\alpha)\left[w^u(2+r)+(1+r)f\right]<f$
  - $\phi(h)=(1-\alpha)w^s>h$

Bequest Dynamics



Bequest Dynamics: Stability of the Steady States



Bequest Dynamics: Steady States and Thresholds

The two stable steady states:

$$\bar{b}^u = \frac{(1-\alpha)w^u(2+r)}{1-(1-\alpha)(1+r)}, \qquad \bar{b}^s = \frac{(1-\alpha)[w^s-(1+r)h]}{1-(1-\alpha)(1+r)}.$$

The unstable steady state:

$$g = \frac{(1-\alpha)[(1+i)h-w^s]}{(1-\alpha)(1+i)-1}.$$

Under the stated assumptions:

$$0 < \bar{b}^u < f < g < h < \bar{b}^s.$$

- f: threshold for investment in education in the current generation.
- g: threshold separating convergence to $\bar{b}^u$ from convergence to $\bar{b}^s$.

Income Distribution and the Long Run Decomposition of the Labor Force

- $\xi_0(b_0)$: density of the initial distribution of inherited wealth
- Long-run fraction of unskilled workers within each generation (Dynasties whose initial inheritance is below g)

$$\lim_{t\to\infty} l_t^u = \int_0^g \xi_0(b_0)db_0 \equiv \bar{l}^u$$

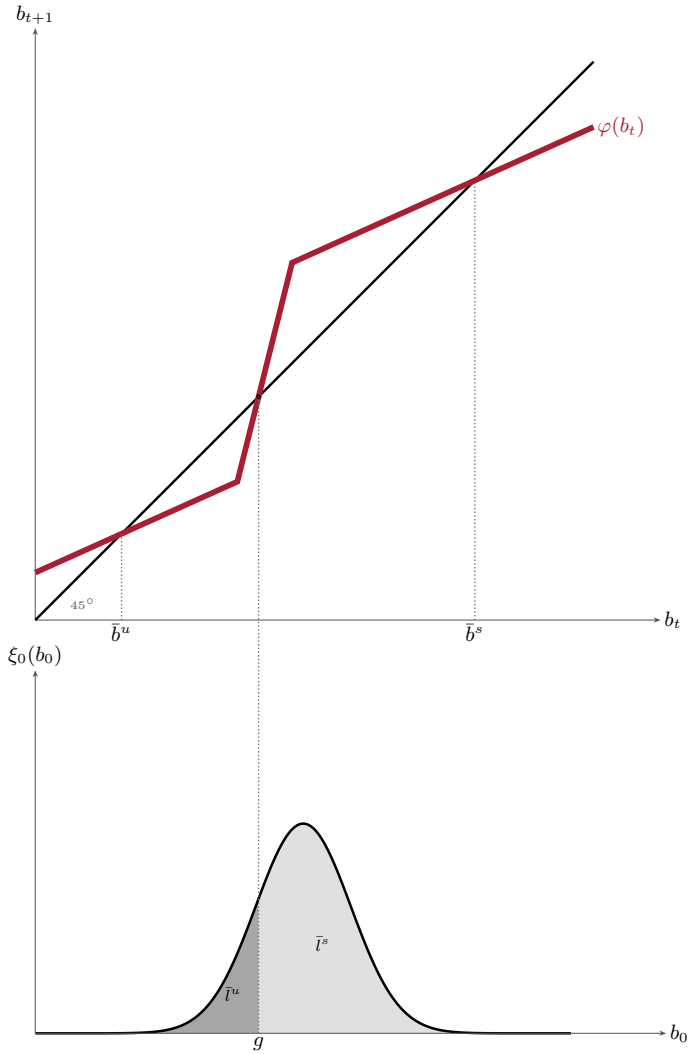
- Long-run fraction of skilled workers within each generation (Dynasties whose initial inheritance is above g)

$$\lim_{t\to\infty} l_t^s = \int_g^\infty \xi_0(b_0)db_0 \equiv \bar{l}^s$$

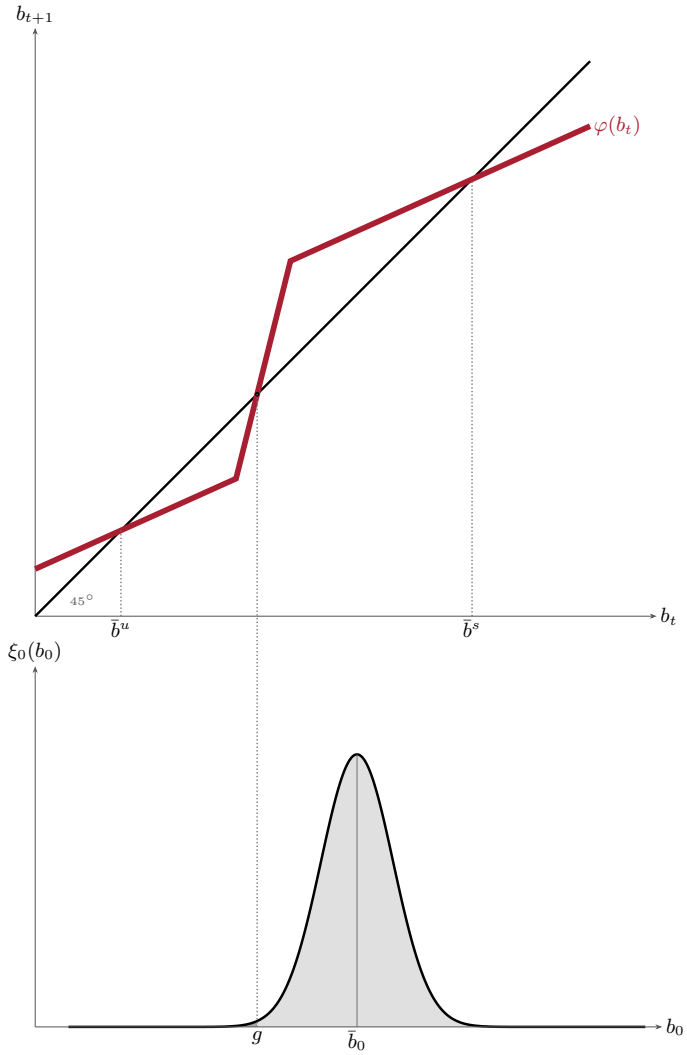
where

$$\frac{\partial \bar{l}^s}{\partial g} = -\xi_0(g) < 0$$

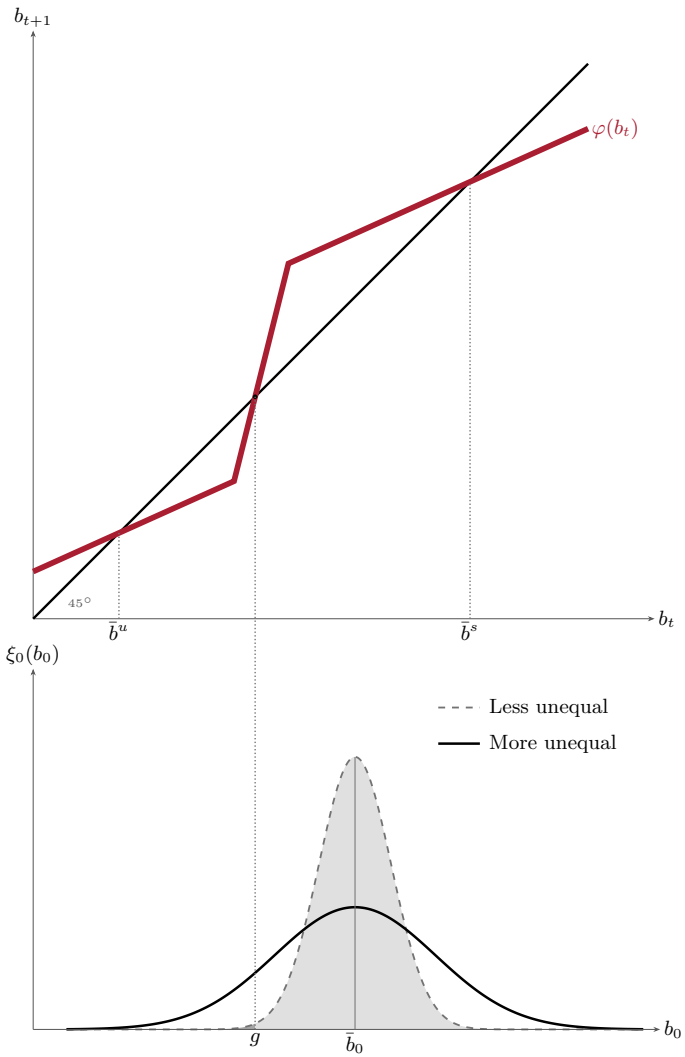
Initial Wealth Distribution and Long-Run Skill Composition



Inequality and Development: Developed Economies (mean inheritance above g)

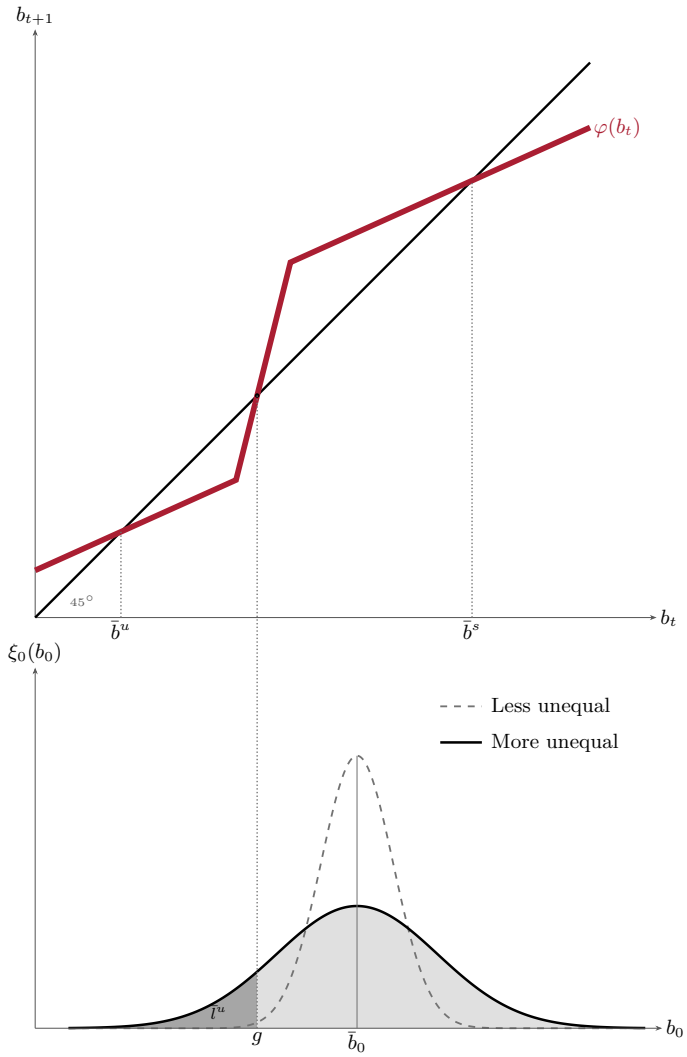


Developed Economies: Lower Human Capital in the Long Run in More Unequal Societies



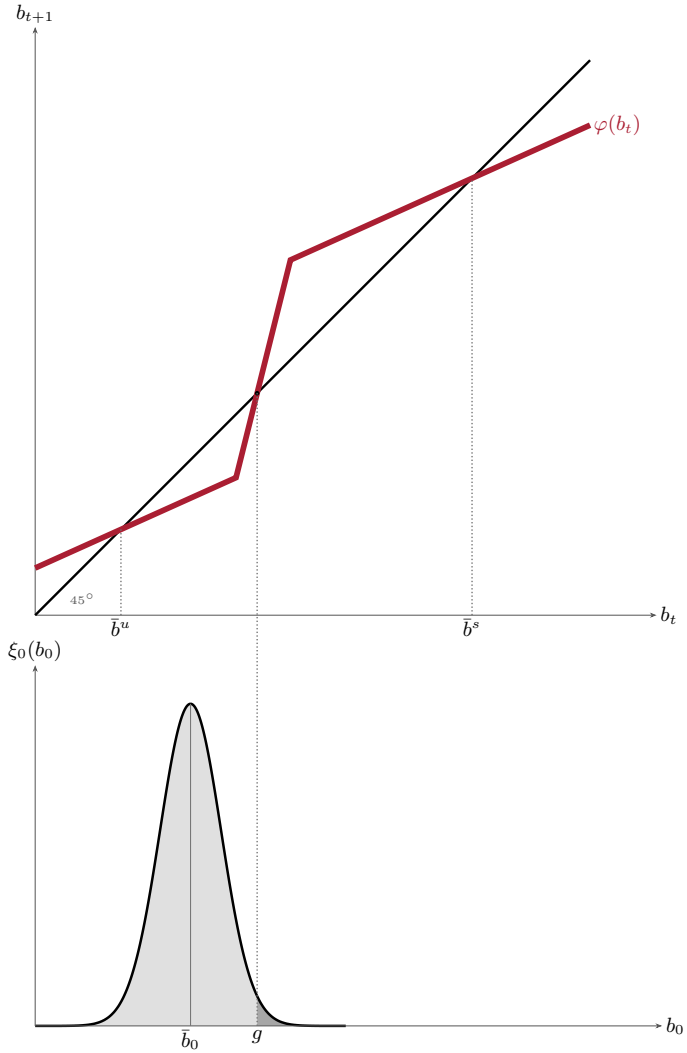
Mean preserving spread of a normal distribution

Developed Economies: Lower Human Capital in the Long Run in More Unequal Societies

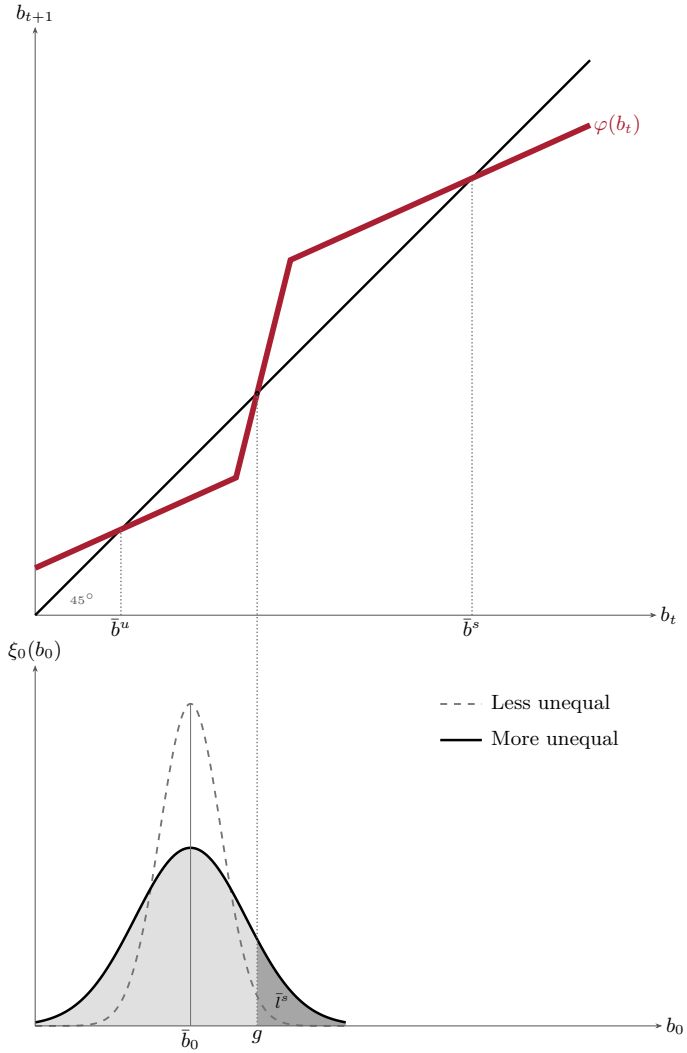


Mean preserving spread of a normal distribution

Inequality and Development: Developing Economies (mean inheritance below g)



Developing Economies: Higher Human Capital in the Long Run in More Unequal Societies



Robustness

The qualitative results are robust to:

- Education cost that is indexed to wages
- Labor-augmenting technical change
- Uncertainty about the outcome of investment in human capital

  provided that wages are endogenous
- Concave production of human capital

  provided that preferences generate a sufficiently convex bequest function

Robustness: Technological Progress and Endogenous Education Cost

Labor-Augmenting Technological Progress: increases the productivity of workers in both the skilled-intensive and the unskilled-intensive sector.

- Production in the skilled-intensive sector

$$Y_t^s = F(K_t, A_t L_t^s) \equiv A_t L_t^s f(k_t); \qquad k_t \equiv K_t/(A_t L_t^s)$$

- Production in the unskilled-intensive sector

$$Y_t^u = A_t a L_t^u$$

- Technological progress

$$A_{t+1} = (1+\lambda)A_t \qquad \lambda > 0.$$

Robustness: Technological Progress and Endogenous Education Cost

Factor Prices

$$w_t^s = A_t[f(k) - f'(k)k] \equiv A_t w^s$$

$$w_t^u = A_t a \equiv A_t w^u$$

$$r_t = r$$

Cost of Education

- Weighted average of the payments to teachers, administrators, and maintenance workers in the school system
- $\Rightarrow$ Weighted average of the wages of skilled and unskilled workers

$$C_t^H = \theta A_t w^s + (1-\theta) A_t w^u \equiv A_t h$$

Wealth: Unskilled Individuals

$$\begin{aligned}\omega_{t+1}^{u} &= (A_t w^u + b_t)(1+r) + A_{t+1} w^u \\ &= A_t w^u (2 + r + \lambda) + (1+r) b_t\end{aligned}$$

Wealth: Skilled Individuals

$$\omega^s_{t+1} = \begin{cases} A_{t+1}w^s - (A_t h - b_t)(1+i) & \text{if} \quad b_t \leq A_t h \\ A_{t+1}w^s + (b_t - A_t h)(1+r) & \text{if} \quad b_t \geq A_t h \end{cases}$$

$$\Longrightarrow$$

$$\omega^s_{t+1} = \begin{cases} A_t\left[w^s(1+\lambda) - (1+i)h\right] + (1+i)b_t & \text{if} \quad b_t \leq A_t h \\ A_t\left[w^s(1+\lambda) - (1+r)h\right] + (1+r)b_t & \text{if} \quad b_t \geq A_t h \end{cases}$$

Education Choice: The Inheritance Threshold

An individual born in period t invests in education if $b_t > f_t$, where

$$f_t = A_t \frac{w^u(2+r+\lambda) - w^s(1+\lambda) + (1+i)h}{i-r}.$$

The threshold in units of technology is constant:

$$\boxed{\widehat{f} \equiv \frac{f_t}{A_t} = \frac{w^u(2+r+\lambda) - w^s(1+\lambda) + (1+i)h}{i-r}}$$

The condition $0 < \widehat{f} < h$ holds if and only if

$$w^s(1+\lambda) - (1+i)h < w^u(2+r+\lambda) < w^s(1+\lambda) - (1+r)h.$$

Bequest Dynamics

Bequests equal a fraction $1-\alpha$ of second-period wealth:

$$b_{t+1} = (1-\alpha)\omega_{t+1}.$$

$$b_{t+1} = (1-\alpha)\begin{cases} A_t w^u(2+r+\lambda) + (1+r)b_t, & 0 \leq b_t < f_t, \\ A_t\left[w^s(1+\lambda) - (1+i)h\right] + (1+i)b_t, & f_t \leq b_t < A_t h, \\ A_t\left[w^s(1+\lambda) - (1+r)h\right] + (1+r)b_t, & b_t \geq A_t h. \end{cases}$$

The three branches correspond to unskilled workers, skilled borrowers, and self-financed skilled workers, respectively.

Bequest Dynamics

Let $\hat{b}_{t+1} \equiv b_{t+1}/A_{t+1}$

$$\hat{b}_{t+1} = \begin{cases} \left[\frac{1-\alpha}{1+\lambda}\right]\{w^u(2+r+\lambda)+(1+r)\hat{b}_t\} & \hat{b}_t \in [0,\hat{f}] \\ \left[\frac{1-\alpha}{1+\lambda}\right]\{[w^s(1+\lambda)-(1+i)h]+(1+i)\hat{b}_t\} & \hat{b}_t \in [\hat{f},h] \\ \left[\frac{1-\alpha}{1+\lambda}\right]\{[w^s(1+\lambda)-(1+r)h]+(1+r)\hat{b}_t\} & \hat{b}_t \in [h,\infty) \end{cases}$$

$\Rightarrow$ Under the conditions stated on the next slide, the normalized dynamical system retains two stable steady states separated by an unstable steady state

Sufficient Conditions for Multiple Steady States

Let $\widehat{\phi}$ denote the normalized bequest map. Given $0 < \widehat{f} < h$, assume:

- $(1-\alpha)(1+r) < 1+\lambda < (1-\alpha)(1+i)$.
- $\widehat{\phi}(\widehat{f}) = \dfrac{1-\alpha}{1+\lambda}\left[w^u(2+r+\lambda) + (1+r)\widehat{f}\right] < \widehat{f}$.
- $\widehat{\phi}(h) = (1-\alpha)w^s > h$.

The normalized system has two stable steady states and an unstable steady state:

$$\boxed{\widehat{g} = \frac{(1-\alpha)\left[(1+i)h - w^s(1+\lambda)\right]}{(1-\alpha)(1+i) - (1+\lambda)}}, \qquad \widehat{f} < \widehat{g} < h.$$

Income Per Capita in the Long Run (Baseline Model)

- Income of a skilled individual in the second period of life (wage and capital income)

$$I_2^s = w^s + (\bar{b}^s - h)r$$

- Income of an unskilled individual in the second period of life (wage and capital income)

$$I_2^u = w^u + (\bar{b}^u + w^u)r$$

- Income of an unskilled individual in the first period of life (only wage income)

$$I_1^u = w^u$$

Income Per Capita in the Long Run

- Aggregate income of residents in the steady state

$$\bar{Y} = I_2^s \bar{l}^s + I_2^u \bar{l}^u + I_1^u \bar{l}^u$$

- Aggregate income (note: $\bar{l}^s + \bar{l}^u = 1$)

$$\begin{aligned} \bar{Y} &= [w^s - rh + r\bar{b}^s]\bar{l}^s + [w^u(2+r) + r\bar{b}^u](1-\bar{l}^s) \\ &= w^u(2+r) + r\bar{b}^u + [(w^s - rh) - w^u(2+r) + r(\bar{b}^s - \bar{b}^u)]\bar{l}^s \end{aligned}$$

- Income per capita

$$\bar{y} = \bar{Y}/2$$

Skill Composition and Income Per Capita in the Long Run

- An increase in the fraction of skilled workers increases income per capita in the steady-state

$$\frac{\partial \bar{y}}{\partial \bar{l}^s} = [(w^s - rh) - w^u(2+r) + r(\bar{b}^s - \bar{b}^u)]/2 > 0$$

since

$$w^s - (1+r)h > w^u(2+r)$$

$$\bar{b}^s > \bar{b}^u$$

- An increase in g reduces income per capita in the steady-state

$$\frac{\partial \bar{y}}{\partial g} = \frac{\partial \bar{y}}{\partial \bar{l}^s}\frac{\partial \bar{l}^s}{\partial g} < 0$$